UNITED STATES PATENT AND TRADEMARK OFFICE

NON-PROVISIONAL PATENT APPLICATION

Conversion from Provisional Application

APPLICATION DATA SHEET

Field	Value
Filing Type	Non-Provisional (Conversion)
Provisional Filed	November 27, 2025
Deadline	November 27, 2026
Entity Status	Micro Entity
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APPLICATION TITLE

UNIVERSAL SMALL COLLECTIBLES RECOGNITION AND MANAGEMENT SYSTEM WITH ENCLOSED ROBOTIC SCANNING CHAMBER, ADAPTIVE INTELLIGENCE, AND CROSS-CATEGORY LEARNING

CLAIM COUNT AND FEES

Category	Count	Fee
Total Claims	71	-
Independent Claims	29	-
Excess Claims (over 20)	51 × $20	$1,020
Excess Independent (over 3)	26 × $42	$1,092
Base Filing Fee	-	$320
TOTAL USPTO		**$2,432**

CLASSIFICATION

- G06K 9/00 - Pattern Recognition
- G06K 9/62 - Methods for recognition

- G06N 3/08 - Neural Networks
- G06Q 30/02 - Marketing, Price Estimation
- B25J 9/16 - Programme-controlled manipulators
- G06T 7/00 - Image analysis
- G06V 30/10 - Optical character recognition

SPECIFICATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims benefit of U.S. Provisional Application filed November 27, 2025, incorporated herein by reference.

FIELD OF THE INVENTION

This invention relates to automated recognition and management systems for collectible items, specifically an integrated hardware and software platform comprising an enclosed scanning chamber with controlled lighting, an articulated robotic arm with camera positioning capability, adaptive optical character recognition, cross-category artificial intelligence learning, and distributed cloud architecture for recognizing, cataloging, grading, and managing small collectible items including trading cards, sports cards, coins, stamps, comic books, action figures, video games, and vinyl records.

BACKGROUND OF THE INVENTION

Technical Field

This invention addresses the $248+ billion global collectibles market through automated recognition and management.

Problems with Prior Art

Fragmented Solutions: TCGPlayer (TCG only), COMC (sports only), Discogs (vinyl only) - each requires separate systems with no data sharing.

Manual Data Entry: 30-60 seconds per item, 5-15% error rates, inconsistent between employees.

Poor Recognition: Mobile apps achieve only 60-85% accuracy due to uncontrolled lighting and variable camera quality.

No Cross-Category Learning: Systems operate in isolation; MTG learning doesn't help Pokemon.

Privacy Concerns: Cloud-only solutions expose business data.

Hardware Limitations: Professional scanners cost $5,000-50,000; consumer solutions limited to phones.

SUMMARY OF THE INVENTION

The invention provides a universal collectibles recognition system comprising:

Hardware Components

1. **Enclosed Scanning Chamber** - Opaque walls, controlled LED lighting (3000K-6500K), positioning surface
2. **Articulated Robotic Arm** - 3+ DOF, end-effector mounted camera, category-specific position profiles
3. **Camera Module** - 12+ megapixels, macro capability, auto-focus
4. **Programmable Lighting** - Addressable LEDs, adjustable intensity/temperature

Software Components

1. **Category Detection** - Automatic item type identification
2. **Multi-Region Extraction** - Category-adaptive region definitions
3. **Bulletproof OCR** - Multi-pass validation, graceful degradation
4. **Cross-Category AI** - Universal pattern learning
5. **Visual Fingerprinting** - Perceptual hashing for matching
6. **Adaptive Shop Personality** - UI customization based on usage

Technical Improvements

- 99%+ recognition accuracy (vs. 60-85% mobile apps)
- 100+ items/hour (vs. 10-15 manual)
- 100% uptime with graceful degradation
- Cross-category learning improving all categories
- Privacy compliance with local data retention

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 - Perspective view of enclosed scanning chamber **FIG. 2** - Cross-sectional view showing interior arrangement **FIG. 3** - Articulated robotic arm assembly detail **FIG. 4** - Lighting system layout (top view) **FIG. 5** - System architecture block diagram **FIG. 6** - Universal scanning method flowchart **FIG. 7** - Multi-region extraction: Trading cards **FIG. 8** - Multi-region extraction: Sports cards **FIG. 9** - Multi-region extraction: Coins **FIG. 10** - Multi-region extraction: Comic books **FIG. 11** - Multi-region extraction: Graded slabs **FIG. 12** - Bulletproof OCR flowchart **FIG. 13** - Cross-category AI learning block diagram **FIG. 14** - Privacy-first distributed architecture **FIG. 15** - AI grading prediction flowchart **FIG. 16** - Robotic position profiles diagram **FIG. 17** - Consumer handheld scanner perspective **FIG. 18** - Barcode-visual cross-validation flowchart **FIG. 19** - Visual fingerprint recognition system **FIG. 20** - Shop intelligence recommendation engine **FIG. 21** - Adaptive UI personalization process

DETAILED DESCRIPTION

1. Enclosed Scanning Chamber (FIGS. 1-2)

The enclosed scanning chamber (100) comprises an opaque housing (101) defining an interior scanning volume (102). Preferred embodiment: repurposed ATX computer case, 12"×12"×18" interior.

Interior surfaces (103) coated with matte black non-reflective material. Access opening (104) with hinged door (105) and latch (106).

Scanning platform (107) with flat upper surface (108) and alignment guides (109) for consistent positioning.

2. Lighting System (FIGS. 2, 4)

Primary light panel (201) with addressable LED elements (204) - WS2812B or compatible, 60 LEDs/meter. Diffusion layer (205) provides even illumination.

Secondary light strips (202) for side-angle "raking light" defect detection.

Lighting controller (206) with Arduino microcontroller enables:

- Independent LED control
- Color temperature: 3000K-6500K
- Intensity: 0-100%
- Category presets
- Synchronized lighting sequences

3. Articulated Robotic Arm (FIGS. 2, 3)

Base mount (301) attached inside chamber. Articulated arm assembly (302) with:

- 3+ rotational joints (303a, 303b, 303c)
- Servo/stepper motors (305)
- Position encoders (306)
- 12" horizontal reach, 8" vertical reach
- ±0.2mm repeatability

End-effector (307) mounts camera module (400).

4. Camera Module (FIG. 3)

Image sensor (401): 64MP (preferred), minimum 12MP Lens system (402): Macro-capable, auto-focus (403) Interface (404): CSI connection to Raspberry Pi

5. Control System (FIG. 5)

Three-node Raspberry Pi 5 cluster:

- **Brok** (192.168.1.169): Camera control, initial capture
- **Snarf** (192.168.1.172): OCR processing
- **Zultan** (192.168.1.152): Orchestration, AI inference

Communication via ZeroMQ messaging.

6. Multi-Region Extraction (FIGS. 7-11)

Trading Cards (FIG. 7):

- Region 601a: Name (top 20%) - Text OCR
- Region 601b: Set Symbol (right side) - Visual match
- Region 601c: Type Line (middle) - Text OCR
- Region 601d: Collector Info (bottom left) - Text OCR
- Region 601e: Artwork (center 50%) - Perceptual hash

Sports Cards (FIG. 8):

- Player Name, Photo, Team, Set, Card Number

Coins (FIG. 9):

- Obverse: Legend, Portrait, Date, Mint Mark
- Reverse: Legend, Design, Denomination

Comics (FIG. 10):

- Cover: Title, Art, Issue#, Publisher, Barcode
- Spine: Title, Issue#, Barcode

Graded Slabs (FIG. 11):

- Company Logo, Grade, Cert Number, Description, Encased Item

7. Bulletproof OCR (FIG. 12)

Auto-path detection searches 10+ installation locations.

Multi-pass validation:

- Pass 1: PSM 6, OEM 3 (standard)
- Pass 2: PSM 7 (single line)
- Pass 3: Aggressive preprocessing (CLAHE, adaptive threshold)

Preprocessing: Grayscale → Threshold → Noise reduction → CLAHE → Deskew → Scale

Graceful degradation: Return empty string (never crash), log failure, request user verification.

8. Cross-Category AI Learning (FIG. 13)

Pattern extraction from transaction data identifies:

- Inventory age effects
- Temporal sales patterns
- Customer segmentation
- Price appreciation patterns

Pattern generalization removes category-specific identifiers.

Cross-category application with confidence modifiers.

Continuous learning updates confidence based on outcomes.

9. Visual Fingerprint Database (FIG. 19)

Hash generation: pHash (DCT) + aHash (mean) + dHash (gradient) = 192-bit fingerprint

Matching: Hamming distance < 10 bits = match

Counterfeit detection via anomaly identification.

Variant detection via fingerprint clustering.

10. Privacy Architecture (FIG. 14)

Local database: Customer info, inventory, pricing, sales (AES-256 encrypted)

Cloud platform: Item identification, market aggregates, model parameters

Synchronization: Only anonymized records uploaded (event ID, category, item ID, timestamp)

Compliance: GDPR, CCPA, PCI DSS compatible

11. AI Grading Prediction (FIG. 15)

Feature extraction:

- Centering (40%): Border width measurement
- Corners (25%): Sharpness/wear detection
- Edges (20%): Straightness/chipping analysis
- Surface (15%): Scratch/defect detection

Grade mapping: 99.5%+ = PSA 10, 95%+ = PSA 9, etc.

Submission recommendation: Calculate ROI vs. grading fee.

12. Consumer Handheld (FIG. 17)

Housing: 5"×3"×1", 8oz, aluminum with rubberized grip Camera: 12MP+, macro lens Ring light: 8 LEDs, adjustable Display: 3.5" touchscreen Processor: ARM Cortex-A53 Memory: 2GB RAM, 16GB storage, 10,000+ item cache Battery: 3000mAh, 8+ hours

Same recognition algorithms as professional units.

13. Robotic Position Profiles (FIG. 16)

Category	Position	X	Y	Z	Angle	Light
Cards	Overhead	0	0	8"	90°	80%
Slabs	Angled	2"	0	10"	45°	60%
Coins	Obverse	0	0	4"	90°	90%
Coins	Raking	3"	0	3"	30°	70%
Comics	Cover	0	0	14"	90°	85%

PRIOR ART ANALYSIS

Comparison Table

Feature	PSA Genamint	TCGPlayer	Manual	NEXUS
Hardware	Phone	None	Loupe	Scanner
Lighting	Ambient	N/A	Variable	Controlled
Categories	Cards	TCG	Any	Universal
Accuracy	85%	N/A	80-90%	99%+
Speed	30/hr	N/A	10-15/hr	100+/hr
Cross-Learn	No	No	No	Yes
Privacy	Cloud	Cloud	N/A	Local

Novel Elements (Not in Any Prior Art)

1. Enclosed scanning chamber with controlled lighting for collectibles
2. Robotic arm with category-specific position profiles
3. Multi-region adaptive extraction protocol
4. Bulletproof OCR with guaranteed uptime
5. Cross-category AI learning
6. Adaptive shop personality
7. Privacy-first distributed architecture
8. Consumer handheld with identical algorithms
9. Barcode-visual cross-validation
10. Universal multi-category grading prediction
11. Visual fingerprint counterfeit detection
12. Cross-category recommendation engine

CLAIMS SUMMARY

System Claims (1-30)

Claim	Subject	Type
1	Enclosed scanning chamber	Independent
2	Articulated robotic arm	Independent
3	Multi-region extraction	Independent
4	Bulletproof OCR	Independent
5	Cross-category AI	Independent
6	Adaptive shop personality	Independent
7	Privacy-first architecture	Independent
8	Consumer handheld	Independent
9	Barcode integration	Independent
10	Grading prediction	Independent
11	Shop intelligence	Independent
12	Visual fingerprint	Independent
13	Universal database	Independent
14	Position profiles	Independent
15	Marketplace integration	Independent
16-30	Dependent claims	Dependent

Method Claims (31-50)

Claim	Subject	Type
31	Universal scanning method	Independent
32	Cross-category learning method	Independent
33	Grading prediction method	Independent
34	Multi-region extraction method	Independent
35	Privacy sync method	Independent
36	Barcode-visual validation	Independent
37	Adaptive UI method	Independent
38	Robotic capture method	Independent

39-50	Dependent claims	Dependent

Apparatus Claims (51-65)

Claim	Subject	Type
51	Scanning chamber apparatus	Independent
52	Handheld scanner apparatus	Independent
53	Robotic arm assembly	Independent
54	Lighting system apparatus	Independent
55-65	Dependent claims	Dependent

AI/ML Claims (66-71)

Claim	Subject	Type
66	Visual fingerprint system	Independent
67	Shop intelligence AI	Independent
68-71	Dependent claims	Dependent

CLAIM DEPENDENCY CHART

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SYSTEM CLAIMS:
1 ─── 16
2 ─── 17
3 ─── 18
4 ─── 19
5 ─── 20
6 ─── 21
7 ─── 22
8 ─── 23
9 ─── 24
10 ── 25
11 ── 26
12 ── 27
13 ── 28
14 ── 29
15 ── 30

METHOD CLAIMS:
31 ─┬─ 39
    └─ 40
32 ─┬─ 41
    └─ 42
33 ─┬─ 43
    └─ 44
34 ─┬─ 45
    └─ 46
35 ── 47
36 ── 48
38 ─┬─ 49
    └─ 50

APPARATUS CLAIMS:
51 ─┬─ 55
    ├─ 56
    ├─ 57
    └─ 58
52 ─┬─ 59
    ├─ 60
    └─ 61
53 ─┬─ 62
    └─ 63
54 ─┬─ 64
    └─ 65

AI CLAIMS:
66 ─┬─ 68
    └─ 69
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67 ─┬─ 70
    └─ 71
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ABSTRACT

A universal collectibles recognition and management system comprising an enclosed scanning chamber housing an articulated robotic arm with integrated high-resolution camera and programmable LED lighting, designed for automated recognition, cataloging, grading prediction, and management of small collectible items.

The system achieves 99%+ recognition accuracy through a category-adaptive multi-region extraction protocol that extracts text via optical character recognition and visual features via perceptual hashing. A bulletproof OCR system with multi-pass validation and graceful degradation ensures 100% uptime regardless of input quality.

Key innovations include: cross-category artificial intelligence learning wherein patterns identified in one collectible category automatically improve recognition and recommendations in all other categories; a privacy-first distributed architecture maintaining sensitive business data locally while sharing only anonymized metrics with cloud services; AI grading prediction achieving 93% accuracy within one professional grade; and adaptive shop personality that customizes the user interface based on observed usage patterns.

The system serves the $248+ billion global collectibles market spanning trading cards, sports cards, coins, stamps, comics, action figures, video games, and vinyl records.

INVENTOR'S DECLARATION

I hereby declare that:

(1) I am the original inventor of the subject matter claimed.

(2) I have reviewed and understand the contents of this application.

(3) The invention has not been publicly disclosed more than one year prior to this filing.

(4) I acknowledge the duty to disclose material information.

(5) I claim benefit of U.S. Provisional Application filed November 27, 2025.

Inventor: Kevin Caracozza **Residence:** Lyndhurst, New Jersey, United States **Address:** 321 Lyndhurst Avenue, Lyndhurst, NJ 07071 **Citizenship:** United States of America

Signature: _____

Date: _____

MICRO ENTITY CERTIFICATION

I hereby certify that I qualify as a micro entity as defined in 37 CFR 1.29(a).

Signature: _____

Date: _____

ATTORNEY INSTRUCTIONS

TO: Lerner David Law Group

This document package contains:

1. Complete specification with detailed description
2. Prior art analysis and differentiation
3. All 71 claims (29 independent, 42 dependent)
4. Claim dependency chart
5. Abstract
6. Declaration and micro entity certification

REQUESTED ACTIONS:

1. Review claims for proper format and antecedent basis
2. Review specification for adequate claim support
3. Prepare formal drawings based on 21 figure descriptions
4. File non-provisional converting from provisional
5. Consider PCT filing (same deadline)

USPTO FEES (Micro Entity): $2,432

FULL MARKDOWN DOCUMENT: E:\NEXUS_USPTO_READY_PATENT_APPLICATION.md (2,585 lines)

Document prepared: January 18, 2026

Status: READY FOR ATTORNEY REVIEW AND FILING